

Network Analysis & Synthesis

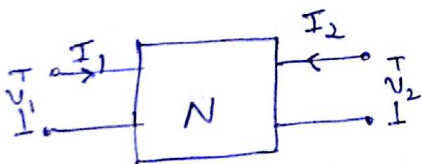
Unit 1 - Signals

Unit 2 - Network Topology & Graph theory Introductory concepts of network graph, cut set, loops, cut set & loop analysis.

Unit 3 - Network theorems, Superposition, Thevenin, Norton, Maximum Power Transfer, reciprocity & Miller theorem.

Unit 4 - Laplace transform, Review of properties & application of Laplace transform of complex waveform & transient waveform response of RLC series, parallel & series-parallel circuits for all excitation

Unit 5 - Two port network: $Z, Y, h, g, ABCD$, inverse $ABCD$ parameters their interconnection & interconnection of two 2 port network, concept of transform impedance



Two port network

Unit 6 - +ve real function, definition & properties, Foster I & II, Cauer I & II form, synthesis of RC, LC & RLC networks, image parameter & basis of two port synthesis.

Books: (1) Von Valkenberg, Network Analysis, PHI

(2) F.F. Kuo "Network Analysis & Synthesis", John Wiley & Sons

(3) Decato & Lin "Linear circuit Analysis" Oxford

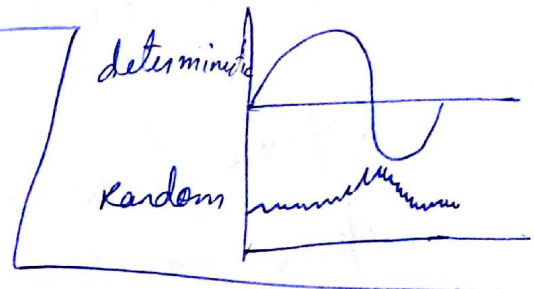
(4) R.C. Dorf.

Unit 1

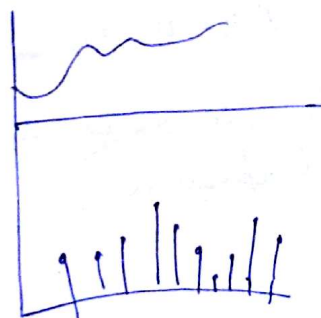
Signal - A function that contains information is signal.
A signal can be either a deterministic or a random signal.

Random signal can take any value over a period of time.

Further classification can be done on the variable present on time.



Continuous or discrete - * On X axis
Continuous over a time, variable on X axis is continuous in nature
Discrete has sample space as variable on X axis is integral in nature
Analog or digital - * On Y axis
Analog can take any value, digital has a fixed value



→ continuous & Analogue

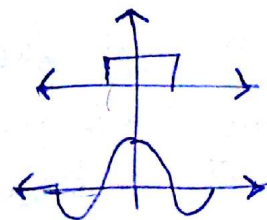
→ discrete & digital

Odd or even function

Even signal is symmetric about Y axis

$$x(t) = x(-t)$$

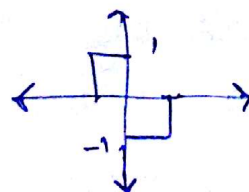
$$y = \cos x$$



Odd signal is anti symmetric about Y axis

$$x(t) = -x(-t)$$

$$y = \sin x$$



$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

Conjugate symmetry & Anti-conjugate symmetry

$$x(t) = a(t) + jb(t)$$

Conjugate symmetry

$$x(t) = x^*(-t)$$

Conjugate anti-symmetry

$$x(t) = -x^*(-t)$$

$$x_{es}(t) = \frac{x(t) + x^*(-t)}{2}$$

$$x_{os}(t) = \frac{x(t) - x^*(-t)}{2}$$

Discrete & continuous signals itself after intervals of time

$$x(t) = x(t \pm nT_0)$$

$$x(n) = x(n+N)$$

$$x(t) = A_0 e^{i\omega t}$$

$$x(t+T_0) = A_0 e^{i\omega(t+T_0)}$$

$$e^{i\omega t} = e^{i\omega t + i\omega T_0}$$

$$1 = e^{i\omega T_0}$$

$$e^{i2\pi k} = e^{i\omega T_0}$$

$$2\pi k = \omega T_0$$

$$T_0 = \frac{2\pi}{\omega}$$

$$[A_0 e^{\theta} = \cos \theta + i \sin \theta]$$

$$\text{now } e^{2\pi k} = \cos 2\pi k + i \sin 2\pi k$$

$$e^{2\pi k} = 1$$

$$x(t) = \sin 2\pi t + \cos 8\pi t$$

$$x_1 = \sin 2\pi t$$

comparing with $\sin 2\pi k t$

$$x(t+T_0) = \sin 2\pi(t+T_0) + \cos 8\pi(t+T_0) \quad \text{u by } k=1 \Rightarrow T_1 = 1$$

$$x_2 = \cos 8\pi t \quad \text{comparing with } \cos 2\pi k t$$

$$k=4 \Rightarrow T_2 = 4$$

$$\text{If } x(t) = x_1(t) + x_2(t)$$

$$\downarrow \quad \quad \quad \hookrightarrow T_1, \omega_2, t_2$$

$$T_1, \omega_1, t_1$$

$$T_0 = \text{LCM}(1, 4)$$

$$T_0 = \text{LCM}[T_1, T_2]$$

$$\omega_0 = \text{HCF}[\omega_1, \omega_2]$$

$$f_0 = \text{HCF}[f_1, f_2]$$

Only possible if

$\frac{T_1}{T_2}$ or $\frac{\omega_1}{\omega_2}$ or $\frac{f_1}{f_2}$ are rational numbers.

$$\cancel{x(t) = \sin 2\pi t = \sin 2\pi(t+T_1)}$$

Energy & Power signals

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt \rightarrow \text{continuous}$$

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 \rightarrow \text{discrete}$$

Continuous

$$\text{Power} = \frac{1}{T_0} \int_{T_0}^{\infty} |x(t)|^2 dt \rightarrow \text{continuous periodic}$$

$$\text{Power} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \rightarrow \text{discrete non-periodic}$$

Discrete

$$P = \frac{1}{N} \sum_{n=N}^{\infty} |x(n)|^2$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=N}^{\infty} |x(n)|^2$$

Points to Ponder

- Energy should be finite & power ^{over a cycle} should be 0
- Energy signals are absolutely integrable signals $\Rightarrow$ it should have finite value. $\int_{-\infty}^{\infty} |x(t)| dt < \infty$
- For power signals power should be finite & energy should be infinite
- Periodic power signals are absolutely integrable over their time period. $\int_{T_0} |x(t)| dt < \infty$

Orthogonal signals

$$\int_{-\infty}^{\infty} x_1(t) \cdot x_2(t) dt = 0 \rightarrow \text{for Aperiodic}$$

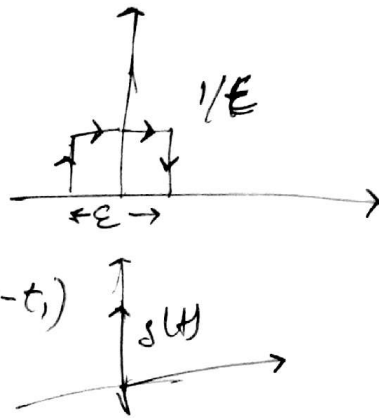
$$\int_{T_0} x_1(t) \cdot x_2(t) dt = 0$$

$\downarrow \qquad \qquad \downarrow$
 $T_1 \qquad \qquad T_2$

Basic Signal

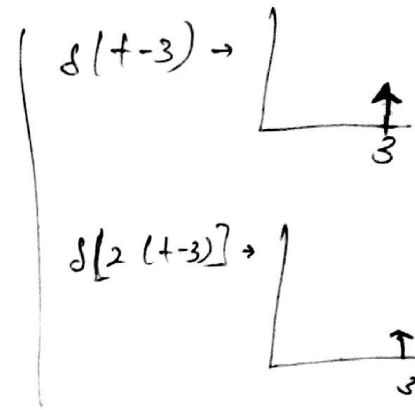
(1) Unit Impulse

$$\delta(t) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \pi(t)$$



$$\rightarrow \delta[a(t-t_1)] = \frac{1}{|a|} \delta(t-t_1)$$

$a \neq 0$



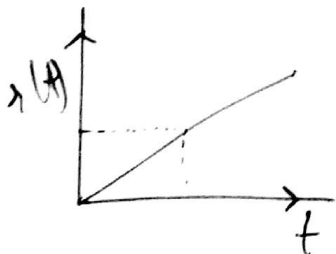
(2) Unit Step Fn

$$u(t) = u(t)$$
$$= 1 \quad t \geq 0$$

$$\frac{du(t)}{dt} = \delta(t) \quad 0 \quad t < 0$$

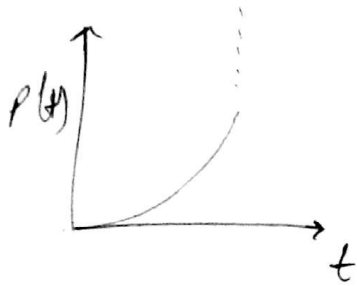
$$u(t) = \int_{-\infty}^t \delta(t) dt$$

(3) Unit ~~to~~ ramp signal



$$r(t) = \int_{-\infty}^t u(t) dt$$
$$= t u(t)$$

(4) Parabolic signal



$$p(t) = \int_{-\infty}^t x(t) dt$$

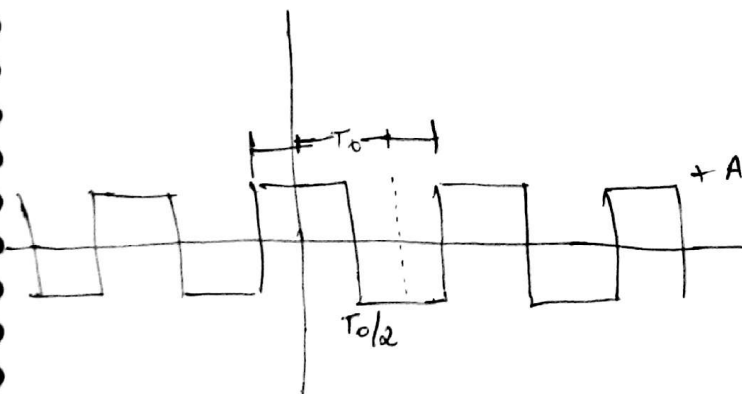
$$= \frac{t^2}{2} u(t)$$

(5) Signum function

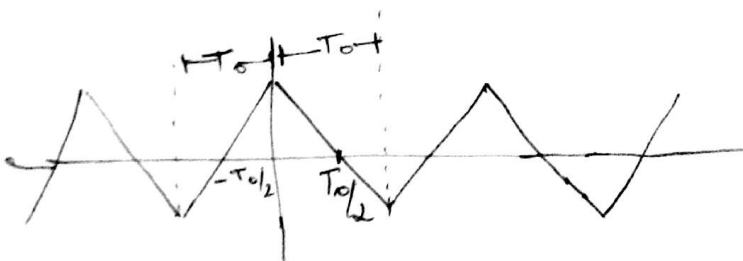
$$\text{Sign}(t) = 1 \quad t > 0$$

$$-1 \quad t < 0$$

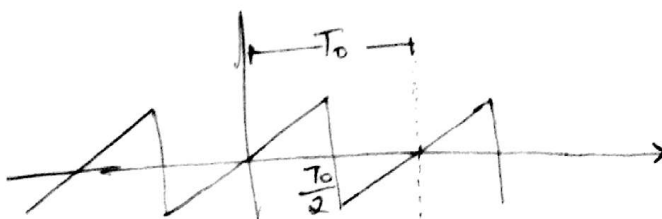
Waveform



Symmetric Square Wave



Symmetric Triangular Wave



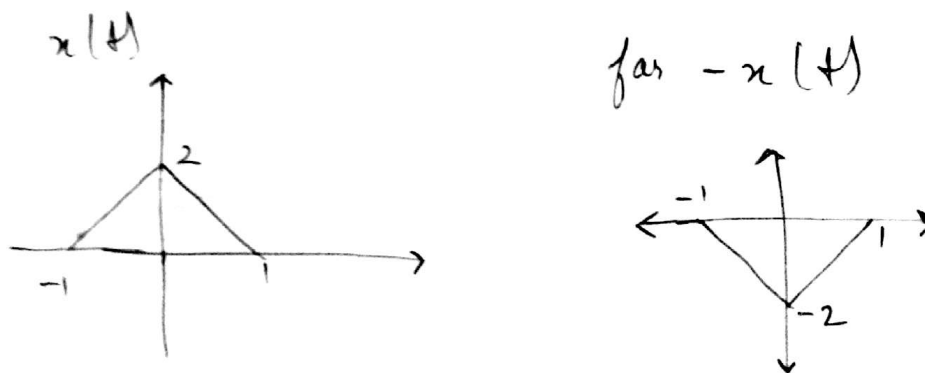
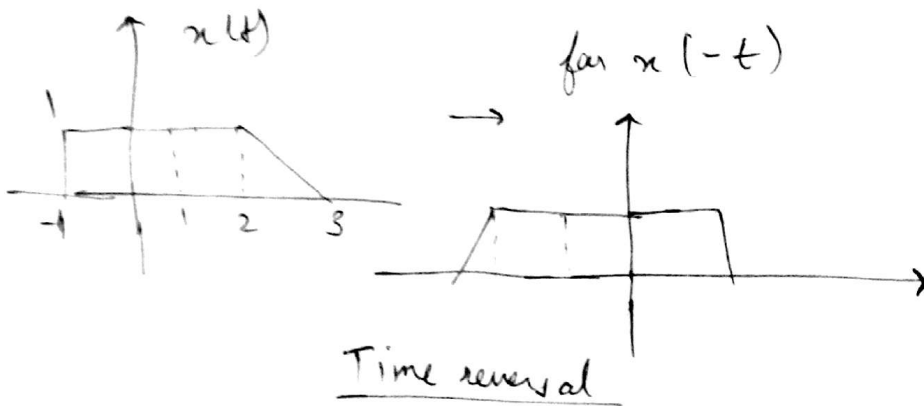
Time scaling

$$x(t) \rightarrow y(t) = x(\alpha t)$$

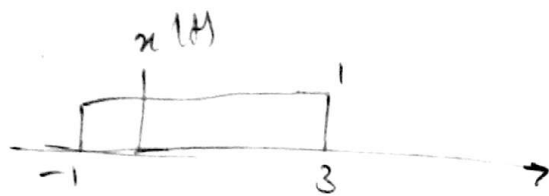
$0 < \alpha < 1 \rightarrow$ expansion of graph
 $\alpha > 1 \rightarrow$ compression

Reversal

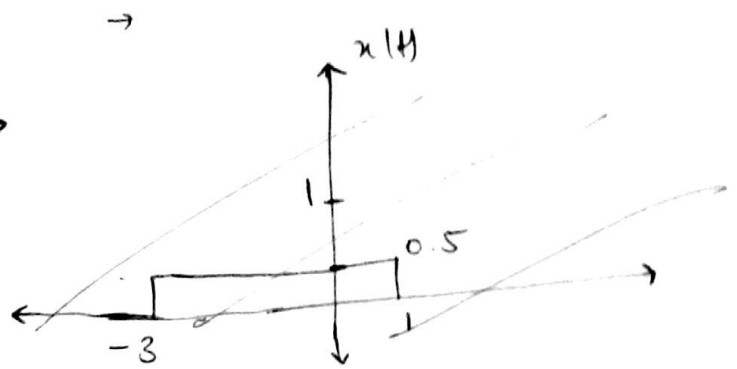
- $\rightarrow$ Time reversal $\rightarrow$ folding of waveform takes along y (axis)
- $\rightarrow$ Amplitude reversal $\rightarrow$ folding " " " " x (axis)



Ex



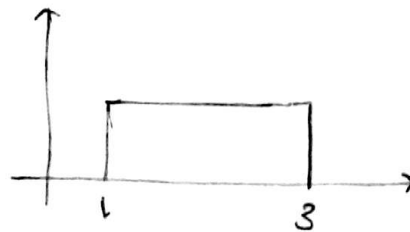
$$x(2t-3)$$



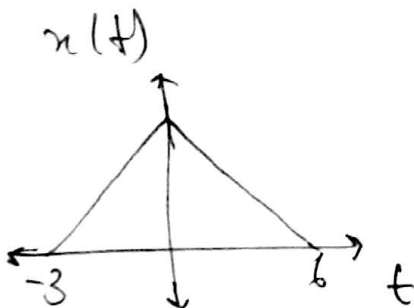
Time scaling
 $x(2t)$



for time shifting
 $x\{2[t-1.5]\}$ so both sides shifted by

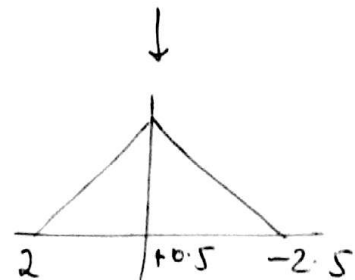
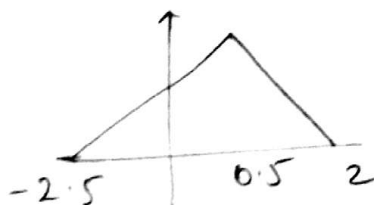
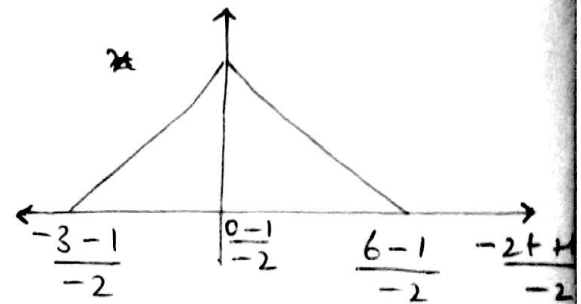


Short cut method



$$y(t) \rightarrow x(t) \rightarrow x(-2t+1)$$

make the variable
 t again



Fourier Series

It is an accurate representation of periodic signal & consists of the sum of sinusoids & the fundamental & harmonic frequencies. Fourier series expansion is used for periodic signal.

$$\Rightarrow f(t) = f(t + nT)$$

$n = \text{integer}$

$T = \text{Time period}$

so Fourier series of $f(t) = \frac{1}{T} \int_T f(t) dt + \sum_{n=1}^{\infty} [a_n \cos n\omega_0 t + b_n \sin n\omega_0 t]$

$\hookrightarrow$ [average value / DC value of $f(t)$]

$$a_0 = \frac{1}{T} \int_T f(t) dt$$

$$a_n = \frac{2}{T} \int_T f(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T} \int_T f(t) \sin n\omega_0 t dt$$

Exponential form of Fourier series

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

* $c_n = c_{-n}$ if $f(t)$ is real function

$$c_n = \frac{1}{T} \int_T f(t) e^{-jn\omega_0 t} dt$$

$$c_0 = \frac{1}{T} \int_T f(t) dt \rightarrow \text{DC value}$$

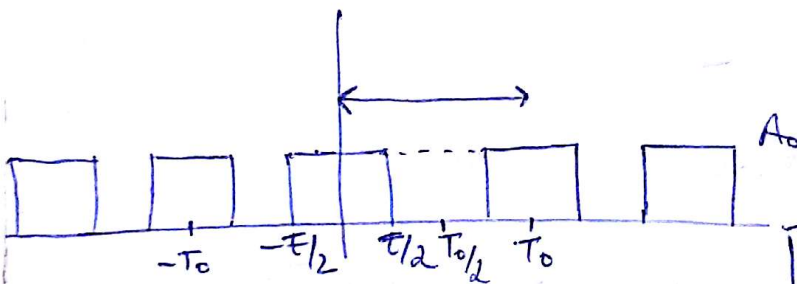
Conditions for existence of fourier series expansion are known as Dirichlet conditions

- ① Signal should be deterministic over its time period i.e it should have finite number of maxima & minima
- ② Signal should have finite number of discontinuities over its time period.
- ③ Signal should be absolutely integrable over its time period.

$$\int_T |f(t)| dt < \infty$$

Example

$f(t) = \text{Magnitude}$



$$C_n = \frac{1}{T_0} \int_{T_0} f(t) e^{-jn\omega_0 t} dt$$

$$= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} A_0 e^{-jn\omega_0 t} dt$$

$$= \frac{A_0}{T_0} \int_{-T/2}^{T/2} e^{-jn\omega_0 t} dt \quad [\because f_n \text{ is limited to } -T/2 \rightarrow T/2]$$

$$= \frac{A_0}{T_0} \left[\frac{e^{-jn\omega_0 t}}{-jn\omega_0} \right]_{-T/2}^{T/2}$$

$$\frac{A_0}{T_0} j n \omega_0 \left[e^{j \frac{n \omega_0 T}{2}} - e^{-j \frac{n \omega_0 T}{2}} \right]$$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$= \frac{A_0}{T_0} \int_{-T/2}^{T/2} (\cos n \omega_0 t + j \sin n \omega_0 t) dt$$

$$\frac{\sin n \omega_0 t}{n \omega_0} \Big|_{-T/2}^{T/2} - j \frac{\cos n \omega_0 t}{n \omega_0} \Big|_{-T/2}^{T/2}$$

$$\frac{A_0}{T_0} \left\{ \left[\frac{\sin n \omega_0 T}{2} - \left(-\sin n \omega_0 T \right) \right] - j \left[\frac{\cos n \omega_0 T}{2} - \frac{\cos n \omega_0 T}{2} \right] \right\}$$

$$\frac{A_0}{T_0} \frac{\sin n \omega_0 T}{n \omega_0}$$

$$\frac{A_0}{T_0} \left(\frac{\sin n \omega_0 T}{n \omega_0} \right)$$

$$T_0 = \frac{2\pi}{\omega_0}$$

$$\frac{2A_0}{2\pi n \omega_0} \frac{\sin n \omega_0 T}{2} = \frac{A_0 \sin(n \omega_0 T)}{n\pi}$$

$$\Rightarrow \frac{A_0}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

$$T = \frac{2\pi}{\omega_0}$$

$$\text{Also } T = 2T$$

$$\omega_0 T = \pi$$

$$C_0 = \frac{1}{T_0} \int_{T_0} f(t) dt$$

$$= \frac{1}{T_0} \int_{-T/2}^{T/2} A_0 dt$$

$$= \frac{A_0}{T_0} T = \frac{A_0 \omega_0}{2\pi} = \frac{A_0 T}{2T} = \frac{A_0}{2}$$

$$[T_0 = 2T \text{ from graph}]$$

Fourier Transform

$$x(t) = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \rightarrow \text{Fourier Inverse}$$

$$X(\omega) = F(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \rightarrow \text{Fourier Transform}$$

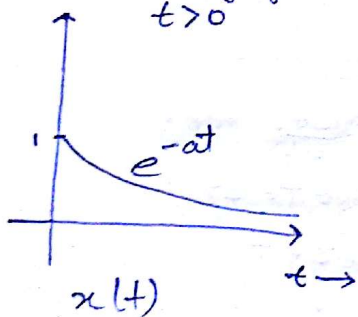
Fourier Transform is used for frequency analysis of signal $x(t)$ is ~~representation~~ referred as F.T of the signal $x(t)$ & $x(t)$ is the representation which is the Fourier inverse transform of $X(\omega)$ as for periodic signal we have representation of a signal as a linear combination of complex exponential, for ~~a~~ periodic signal we have same representation in terms of $X(\omega)$

Conditions for Existence of Fourier Transform

Dirichlet conditions are sufficient condition for conversion of Fourier Transform.

Q $x(t) = e^{-at} u(t)$, find Fourier transform $X(\omega)$ and also plot the magnitude & phase plot of the same.

$u(t)$ signifies $t > 0$



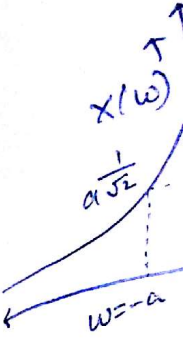
$$X(\omega) = \int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \Big|_0^{\infty} = - \left[0 - \frac{1}{a+j\omega} \right]$$

$$X(\omega) = \frac{1}{a+j\omega}$$

$$|X(\omega)| = \frac{1}{\sqrt{a^2}}$$



Linear Ti

$$x(t) \rightarrow h(t)$$

$$c_1 x(t) \rightarrow h(t)$$

$$x(t) + y(t) \rightarrow$$

A system proportional

$$c_1 e_1(t)$$

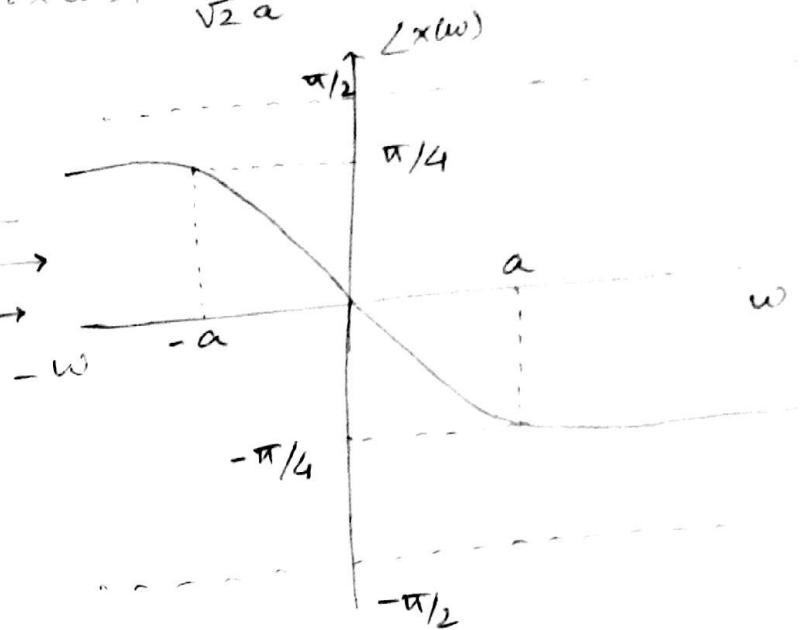
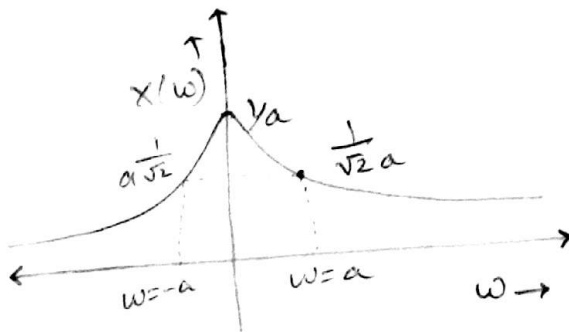
$$c_2 e_2(t)$$

$$c_1 e_1(t) + c_2 e_2(t)$$

$$|x(w)| = \frac{1}{\sqrt{a^2 + w^2}}$$

$$\angle x(w) = -\tan^{-1}\left(\frac{w}{a}\right) \quad \because \text{a } \tan^{-1} \text{ has limit from } +\pi/2 \rightarrow -\pi/2$$

when $w = a$
 $|x(w)| = \frac{1}{\sqrt{2}a}$



Linear Time Invariant (LTI)

$$x(t) \rightarrow [h(t)] \rightarrow y(t) = h(t) \cdot x(t)$$

$$C x(t) \rightarrow [h(t)] \rightarrow y(t) = C x(t) \cdot h(t)$$

$$x(t) + y(t) \rightarrow [h(t)] \rightarrow w(t) = [x(t) + y(t)] h(t)$$

Properties of Linear system

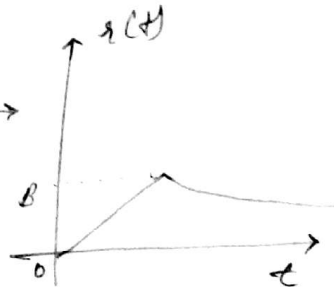
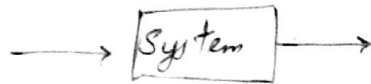
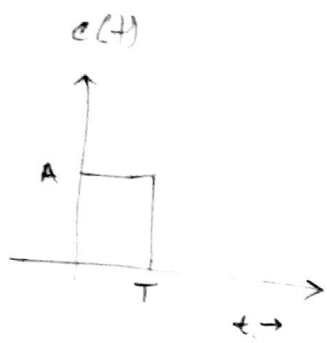
A system is linear if the principle of superposition & principle of proportionality holds.

$$C_1 e_1(t) \rightarrow \text{System} \rightarrow C_1 r_1(t)$$

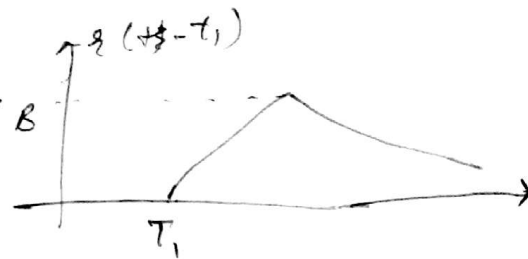
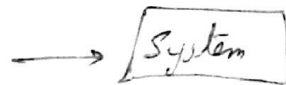
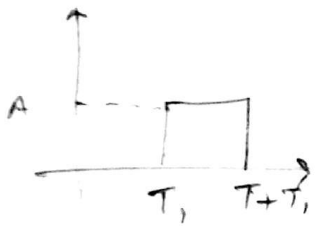
$$C_2 e_2(t) \rightarrow \text{System} \rightarrow C_2 r_2(t)$$

$$C_1 e_1(t) + C_2 e_2(t) \rightarrow \text{System} \rightarrow C_1 r_1(t) + C_2 r_2(t)$$

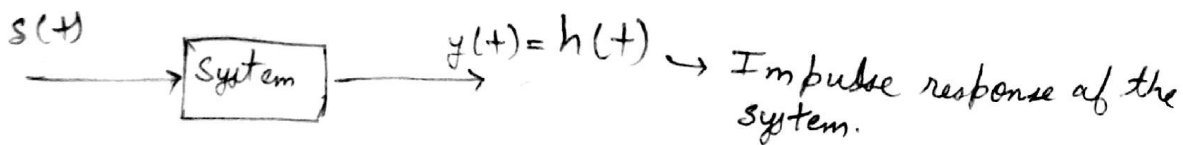
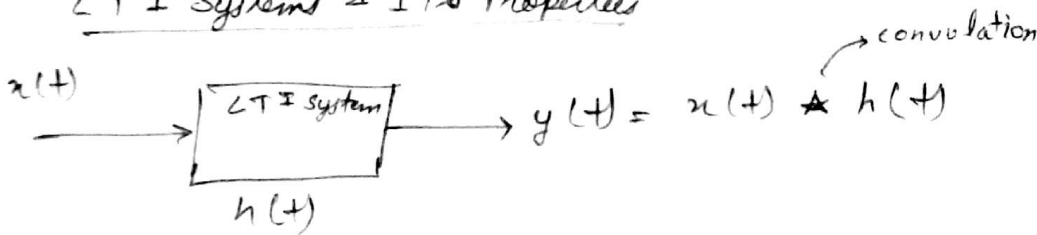
EDC
Lab 517



$e(t-t_1)$



LTI Systems & Its Properties



Here $h(t)$ is the impulse response of the system,

$H(s)$ or $H(\omega)$ is known as Transfer fⁿ of the system

A system is linear i.e it holds principle of super position & proportionality and time invariance i.e any delay provided in the input must reflect in the output.

Output of LTI System is given by convolution of the input with the impulse response of the system.

Convolution is the mathematical tool to represent the output of an LTI system

First order circuit = R-C or RL, having only one energy storing element. (C or L)

for discrete signal

$$y(n) = x(n) * h(n)$$

$$= \sum_{k=-\infty}^{\infty} h(k) x(n-k)$$

Transfer f^n is defined as ratio of Laplace Transform of output to the Lap. Transform of Input when all I.C. = 0

$$H(s) = \frac{Y(s)}{X(s)} \quad \left| \begin{array}{l} \text{zero initial condition} \end{array} \right.$$

Total response of the system = $ZIR + ZSR \rightarrow$ zero state response
 $\downarrow$ $\downarrow$
 zero input response due to applied input
 due to Initial conditions

Properties of LTI system

① Commutative Property

$$x(t) * h(t) = h(t) * x(t)$$

② Distributive Property

$$x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + h_2 x(t) * h_2(t)$$

③ Associative Property

$$x(t) * [h_1(t) * h_2(t)] = [x(t) * h_1(t)] * h_2(t)$$

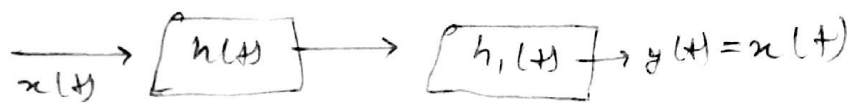
Stability of LTI system

$$\int_{-\infty}^{+\infty} |h(t)| dt < \infty$$

Causality of system

$$h(t) = 0, t < 0$$

Invertibility of LTI system

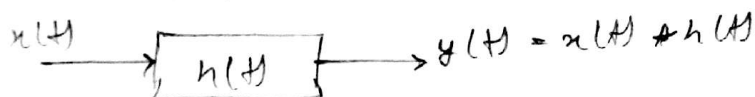


$$h(t) * h(t) = \delta(t)$$

Step Response of LTI

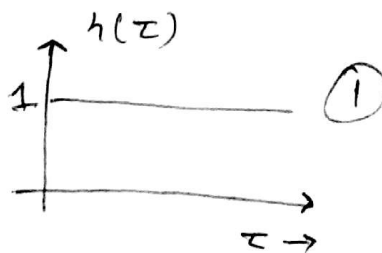
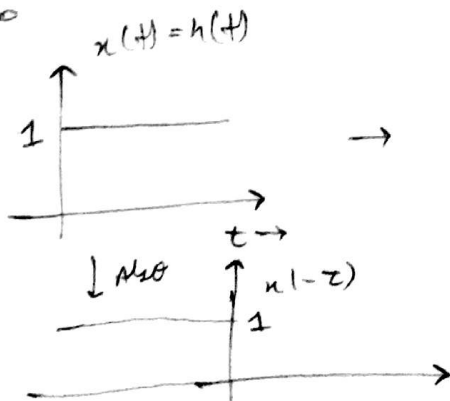
$$s(t) = \int_{-\infty}^{+\infty} h(t) dt$$

$$h(t) = \frac{d s(t)}{dt}$$



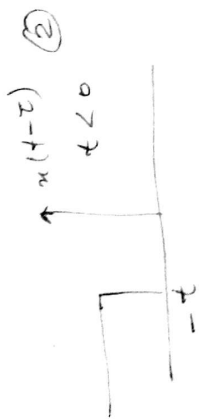
$$y(t) = x(t) * h(t)$$

$$= \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$



$$\begin{aligned} x(t) &= u(t) \\ h(t) &= u(t) \\ y(t) &= ? \end{aligned}$$

Now shifting by t of $x(t-z)$



① $x(2) = 0$

① $x(3) = 1$ also range will be of $0 \rightarrow t$

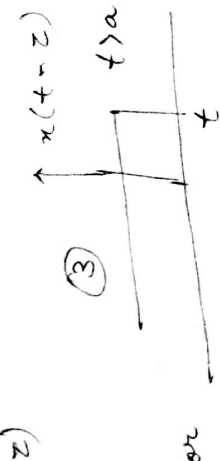
$$\int_0^t 1 \cdot 1 \cdot dt \quad y(t) = t u(t)$$

As $\mathcal{L}\{x(t)\} = \mathcal{L}\{x(t)\} = \frac{1}{s^2}$

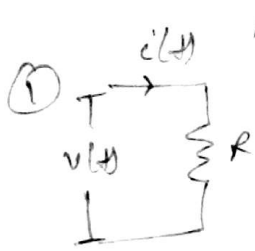
$\mathcal{L}\{y(t)\} = Y(s) = \frac{1}{s^3}$

$x(t) * h(t) = X(s) \cdot Y(s) = \frac{1}{s^2}$

$\mathcal{L}^{-1}\{X(s) \cdot Y(s)\} = \mathcal{L}^{-1}\{\frac{1}{s^2}\} = t u(t) = y(t)$



Basis of Network Theory



$$v(t) = i(t) R$$

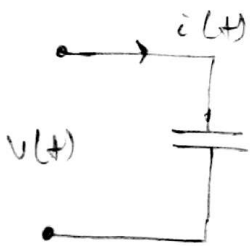
$$V(s) = I(s) R$$

$$V = |V| \angle \theta$$

$$I = |I| \angle \phi$$

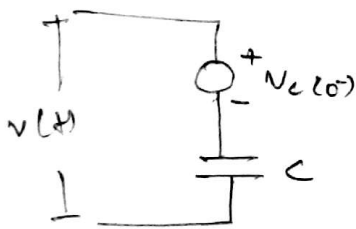
$$\theta = \phi$$

② Capacitor



$$i(t) = C \frac{dv(t)}{dt}$$

$$v(t) = \frac{1}{C} \int_0^t i(t) dt + \underset{\text{initial condition}}{v_c(0^-)}$$



$$V(s) = \frac{1}{Cs} I(s) \rightarrow \text{Laplace}$$

$$V = \frac{1}{j\omega C} I \rightarrow \text{Phasor}$$

$$\boxed{Z_C = \frac{1}{sC}} \rightarrow \text{Impedance}$$

* Voltage across capacitor does not change instantaneously, if $i(t)$ does not contain Impulse or derivative of Impulse

$$\Rightarrow v_c(0^-) = v_c(0^+)$$

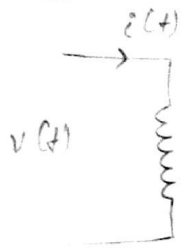
at $t = 0, s = \infty$

$C \Rightarrow$ short circuit

at $t = \infty, s = 0$

$C \Rightarrow$ open circuit

③ Inductor



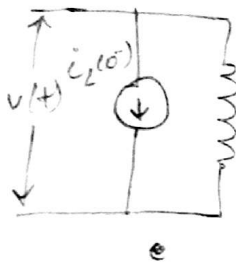
$$v(t) = L \frac{di}{dt}$$

$$i(t) = \frac{1}{L} \int_0^t v(t) dt + i_L(0^-)$$

$$v(s) = sL I(s)$$

$$Z_L = sL$$

$$V = j\omega L I$$



$i_L(0^-) = i_L(0^+)$ iff when $v(t)$ does not have a function of impulse or derivative of impulse

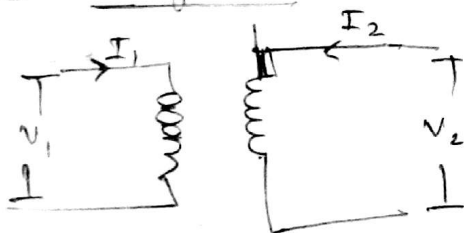
$$\text{At } t=0, s=\infty$$

$$L \Rightarrow 0 \cdot C$$

$$t=\infty, s=0$$

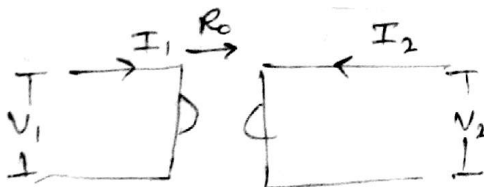
$$L \Rightarrow s \cdot C$$

④ Transformer



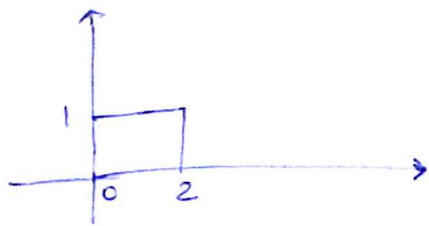
$$\frac{V_1}{V_2} = \frac{N_1}{N_2}; \quad \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

⑤ Gyration (Converts b/w capacitive $\rightleftharpoons$ inductive)



$$V_1 = +R_0 I_2$$

$$V_2 = -R_0 I_1$$



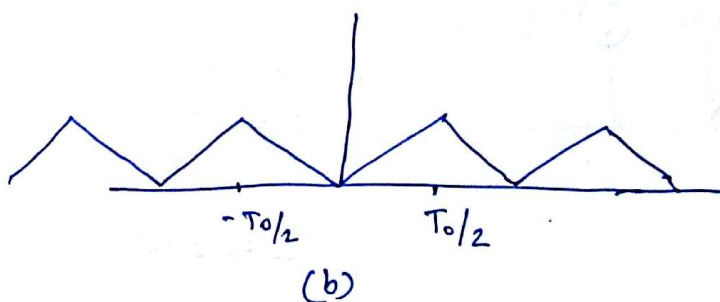
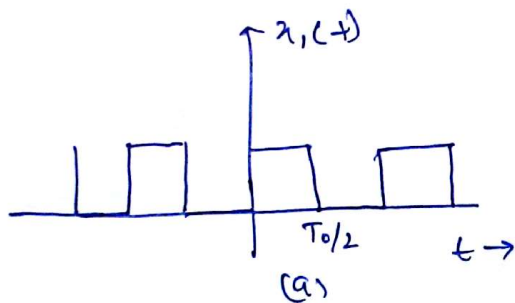
To make this signal
 $u(t) - u(t-2)$

Q1 Find energy of signal $x(t) = e^{-at} u(t)$; $a > 0$

Q2 Find energy of signal $x(t) = e^{-a|t|}$, $a > 0$

$$\begin{matrix} -\infty \rightarrow 0 & e^{at} \\ 0 \rightarrow \infty & e^{-at} \end{matrix}$$

Q3 Find Power of signal $x_1(t)$ & $x_2(t)$ given as



Q4 Find Power of signal $x(t) = A_0 \sin \omega_0 t$

$$\begin{aligned} \text{Q1 } E &= \int_0^{\infty} e^{-2at} dt \\ &= \left. \frac{e^{-2at}}{-2a} \right|_0^{\infty} \\ &= 0 + \frac{1}{2a} = \frac{1}{2a} \end{aligned}$$

$$\text{Q2 } E = \int_{-\infty}^0 e^{at} dt + \int_0^{\infty} e^{-at} dt$$

$$\begin{aligned} E &= \left[\frac{e^{at}}{a} \right]_{-\infty}^0 + \left[-\frac{e^{-at}}{a} \right]_0^{\infty} \\ &= \left(\frac{1}{a} - 0 \right) + \left(0 - \left(-\frac{1}{a} \right) \right) \\ &= \frac{1}{a} + \frac{1}{a} \\ &= \frac{2}{a} \end{aligned}$$

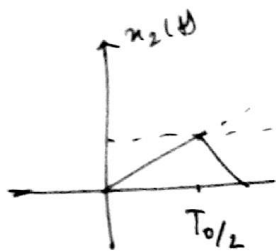
$$\begin{aligned} &= \frac{1}{a} + \frac{1}{a} \\ &= \frac{2}{a} \end{aligned}$$

Q3

$$P = \frac{1}{T_0} \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \frac{2}{T_0} \int_{-\infty}^{\infty} A_0^2 dt = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} A_0^2 dt$$

$$= \frac{2A_0^2}{T_0} \left(\frac{T_0}{2} - \frac{-T_0}{2} \right) = \frac{A_0^2}{T_0} \left(\frac{T_0}{2} + \frac{T_0}{2} \right) = \frac{A_0^2}{T_0} \cdot T_0 = A_0^2$$



$y = mx$ ($\because$ a st. line passing thru origin)

$$y = mx \rightarrow x_2(t) = \frac{2A_0}{T_0} t$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2A_0}{T_0}$$

$$(x_2(t))^2 = \frac{4A_0^2}{T_0^2} t^2$$

$$P = \frac{1}{T_0} \int_0^{T_0/2} \frac{4A_0^2}{T_0^2} t^2 dt$$

$$= \frac{1}{3T_0^3} \left[t^3 \right]_0^{T_0/2}$$

$$= \frac{8A_0^2}{3T_0^3} \cdot \frac{T_0^3}{8}$$

$$= \frac{A_0^2}{3}$$

Q4 $x(t) = A_0 \sin \omega_0 t$

$$P = \frac{1}{T_0} \int_0^{T_0} A_0^2 dt$$

$$= \frac{1}{T_0} \int_0^{\pi} A_0 \sin \omega_0 t dt$$

$$= \frac{A_0}{T_0} \int_0^{\pi} \sin \omega_0 t dt$$

$$= \frac{2A_0}{T_0} \int_0^{\pi/2} \sin \omega_0 t dt$$

$$= -\frac{2A_0}{T_0} \frac{\cos \omega_0 t}{\omega_0} \Big|_0^{\pi/2}$$

$$= \frac{2A_0}{T_0}$$

Q

~~2x4~~

$r(t) = \text{ramp signal}$

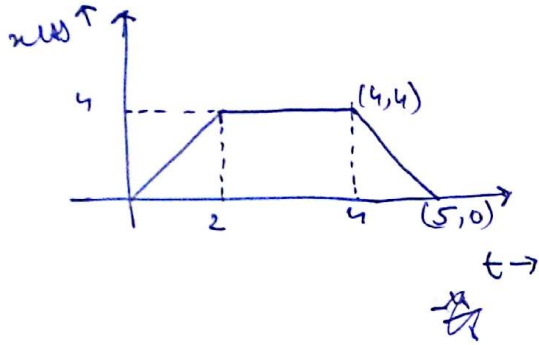
slope from 0 to 2

To make it st to dec

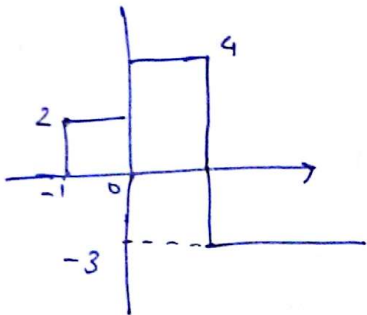
$$2r(t) - 2r(t-2) - 4r(t-4) + 4r(t-5)$$

$\hookrightarrow$ To make it st line
 $\hookrightarrow$ To make it st line

(1)



(2)

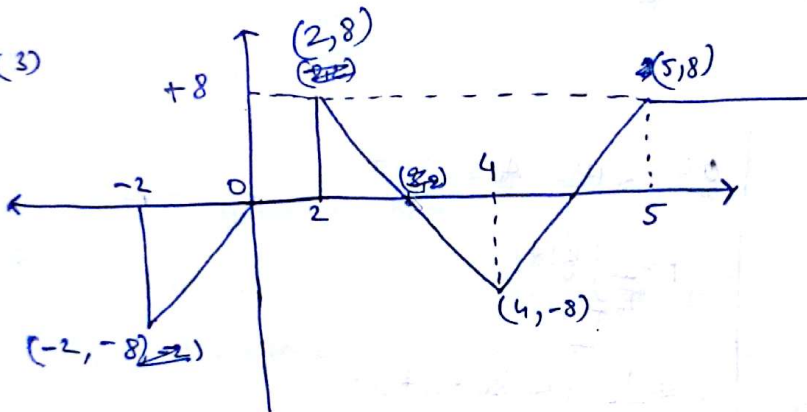


$$2u(t+1) + 2u(t) - 7u(t-1)$$

$$\frac{-8+2}{4} = \frac{-10}{4} = -2.5$$

$$\frac{-8-8}{2} = -8$$

(3)



Kell at v

$$\frac{V_b - V_c}{5}$$

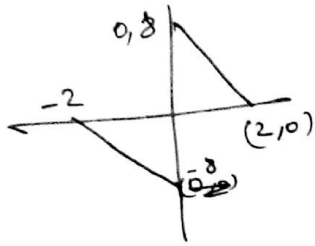
Kell at

$$V_c = V_b$$

$$-8u(t+2) + [4r(t+2)] + 4r(t) + u(t-2) - 8r(t-2) + 16r(t-4) - 16r(t-5)$$

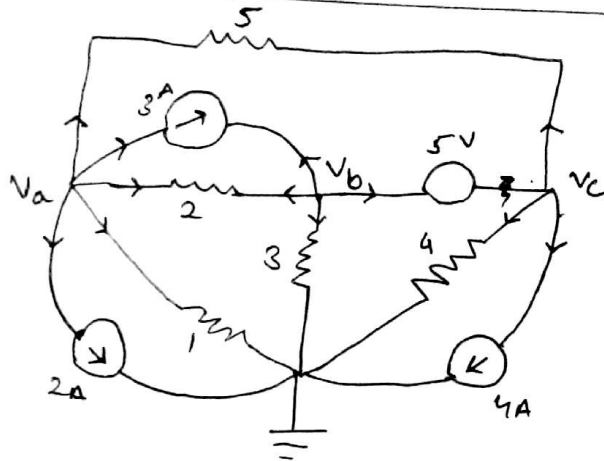
$$-8u(t+2) + 4r(t+2) - 4r(t) + u(t-2) - 8r(t-2) + 16r(t-4) - 16r(t-5)$$

$$or \frac{-8}{-2} = 4$$



$$-4s(t+2) - 2s(t+2) + 16u(t) - 4s(t) + 4s(t-2)$$

$$-4s(t+2) + 4s(t) + 16u(t) - 4s(t) + 4s(t-2)$$



KCL at V_A

$$2 + \frac{V_A - 0}{1} + \frac{V_A - V_B}{2} + 3 + \frac{V_A - V_C}{5} = 0$$

KCL at V_B

$$\frac{V_B - V_C}{5} - 3 + \frac{V_B - V_A}{2} + \frac{V_B - 0}{3} + \frac{V_C - 0}{4} + 4 + \frac{V_C - V_A}{5} = 0$$

by V_C

KCL at V_C

$$V_C = V_B - 5$$

Circuit Transient Analysis of 1 order system

Circuits that contain capacitor & Inductor can be rep. by diff eqⁿ.
the order of diff eqⁿ is usually = to the number of Cap + Inductor in circuit.

Circuits that contain only one capacitor & no inductor can be rep by 1st order diff eqⁿ.

Like wise if only one cap. & no Ind. is present the eqⁿ is still 1 order only.

In general the complete response of first order circuit is the rep that consist of sum of two parts, one is the natural response & the other is forced response.

Complete response = natural response + forced response.

natural response is when Input is said to 0 & forced response is particular solⁿ of diff eqⁿ.

The complete response of first order circuit will be depending on an I.C usually a capacitor voltage or an inductor current at a particular time.

let t_0 denote the time at which I.C is given, the natural response of the first order circuit will be the form

$$x_n = K e^{-(t-t_0)/T}$$

natural response $t_0 = 0$

$$\Rightarrow x_n = K e^{-t_0/T}$$

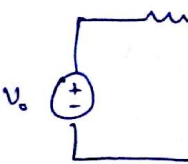
Here K depends upon I.C

→ In the cases when I/P is constant or sinusoidal, the first response is steady state response and natural response is called transient response.

Steps to

- (1) Find the response at time t
- (2) Find the response at time t_0
- (3) Add the natural response & forced response

Response



$$V_0 = IR$$

$$V_0 = IR$$

$$V_0$$

$$K = \frac{V_0}{R + \frac{1}{sC}}$$

Both are

$$\frac{d}{dt} x(t)$$

$$\Rightarrow \frac{d}{dt} x(t)$$

$$T$$

Steps to get complete response of second

(1) Find the first response before the disturbance evaluate this response at time $t = 0$ to obtain the i.c of the energy storage element.

Find the first response after the disturbance

add the natural response $x_n = t e^{-t/T}$ to the first response to get the complete response and use $t = 0$ to evaluate the value of constant K

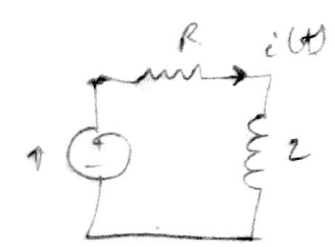
Assume the input voltage is a constant I/P



$$\frac{dV}{dt} = C \frac{dV}{dt}$$

$$P \quad L(t) = C \frac{dV}{dt}$$

$$L(t) = C \frac{dV}{dt}$$



$$-V_0 + Ri(t) + L \frac{di(t)}{dt} = 0$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{V_0}{L} K$$

①
②

$$V_0 - i(t)R - V(t) = 0$$

$$V_0 - i(t)R - V(t) = 0$$

$$\frac{di(t)}{dt} = \frac{1}{L} (V_0 - i(t)R)$$

(1)

Let us assume the form of the function

$i(t) = A e^{-t/T} + B$ where R & L are constant

$i(t) = A e^{-t/T} + B$ where $i(t)$ is first response.

Variable separating & integrating this gen. (A) with

$$\frac{d}{dt}x + \frac{x}{\tau} = K$$

$$\frac{dx}{dt} = \frac{K\tau - x}{\tau}$$

~~Int by~~
$$\frac{dx}{K\tau - x} = \frac{dt}{\tau}$$

Int by

$$\ln(x - K\tau) = \frac{-t}{\tau} + C$$

$$x - K\tau = e^{-t/\tau} \cdot e^D$$

$$x(t) = K\tau + Ae^{-t/\tau} \quad \text{--- (i)}$$

where $A = e^D$ is determined from I.C $x(0)$

~~where~~ let $t=0$

$$x(0) = K\tau + A$$

$$\boxed{A = x(0) - K\tau}$$

$$x(t) = K\tau + (x(0) - K\tau)e^{-t/\tau}$$

$t \rightarrow \infty$ (After a very long time)

$$x(\infty) = K\tau$$

$$x(t) = x(\infty) + [x(0) - x(\infty)]e^{-t/\tau}$$

$\uparrow$
I.C

$\uparrow$
F.C

diff ① w.r.t x

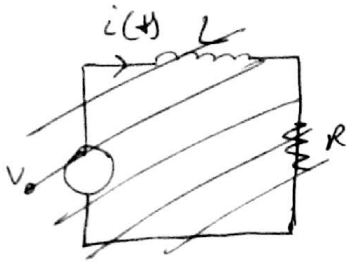
$$x'(t) = 0 - \frac{1}{\tau} [x(0) - x(\infty)]e^{-t/\tau}$$

let $t=0$

$$x'(0) = -\frac{1}{\tau} [x(0) - x(\infty)]$$

$$\boxed{\tau = \frac{x(\infty) - x(0)}{\left. \frac{d}{dt}x(t) \right|_{t=0}}}$$

~~For inductor~~



$\therefore$ for ~~inductor~~ capacitor

$$\frac{dV(t)}{dt} + \frac{1}{RC} V(t) = \frac{V_0}{RC}$$

$$Z = RC$$

$$K = \frac{V_0}{RC}$$

$$A = V(0) - KZ$$

$$= V(0) - \frac{V_0}{RC} RC$$

$$A = V(0) - V_0$$

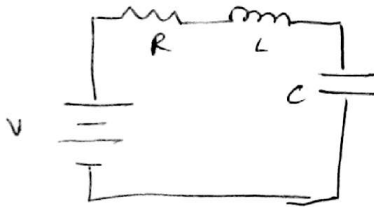
$$x_n = A e^{-(t-t_0)/Z}$$

$$= [V(0) - V_0] e^{-(t-t_0)/RC}$$

$$V = 10V \quad R = 10\Omega \quad L = 1H \quad C = 10\mu F$$

$$i(0+), i'(0+), i''(0+)$$

$$V_L(0) = 0$$

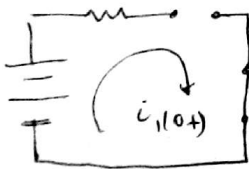


$$\frac{dZ}{dt} = CV$$

$$Z = CV$$

$$i = C \frac{dv}{dt}$$

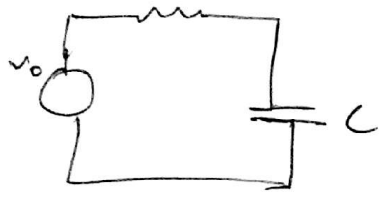
$$V = iR + L \frac{di}{dt} + \frac{1}{C} \int i dt$$



$$V(\underline{t}) = L \frac{di(0+)}{dt} + \frac{1}{C} \int i(0+) dt + i(0+) R$$

$$= \frac{Z}{C} + iR$$

RC



$$\frac{dV(t)}{dt} + \frac{1}{RC} V(t) = \frac{V_0}{RC}$$

$$x(t) = V(t), \quad \tau = RC$$

$$K = \frac{V_0}{RC}$$

$$x(t) = x(\infty) + [x(0) - x(\infty)] e^{-t/\tau}$$
$$= K\tau + \left[\frac{V(0)}{K} + \cancel{K\tau} - K\tau \right] e^{-t/\tau}$$

$$V(t) = \frac{V_0}{RC} \cdot RC + \left[V(0) - \frac{V_0}{RC} \cdot RC \right] e^{-t/RC}$$

Initial condition

$$V(t) = V_0 + [V(0) - V_0] e^{-t/RC}$$

$$V(t) = \underbrace{V_0}_{\text{Forced Response}} + \underbrace{[V(0) - V_0] e^{-t/\tau}}_{\text{Natural response}}$$

$\therefore$ Second component of eqⁿ dies out with an $\uparrow$ of time t this is known as transient or natural response of the circuit.

Whereas 1st component of eqⁿ is the forced response of the system $\therefore$ complete response of the system is given by (forced + natural)

RL

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{V_0}{L}$$

$$\tau = L/R \quad K = \frac{V_0}{L}$$

$$x(t) = x(\infty) + [x(0) - x(\infty)] e^{-t/\tau}$$

$$i(t) = \frac{V_0}{L} \cdot \frac{L}{R} + \left[i(0) - \frac{V_0}{R} \right] e^{-t/\tau}$$

$$= \underbrace{\frac{V_0}{R}}_{\text{Forced response}} + \underbrace{\left[i(0) - \frac{V_0}{R} \right]}_{\text{Initial condition}} e^{-t/\tau}$$

Natural response

→ Stability of I order eqⁿ / circuit

$$x_n(t) = K e^{-t/\tau}$$

$$x(t) = x_n(t) + x_f(t)$$

When τ is +ve the natural response vanishes as $t \rightarrow \infty \Rightarrow x_n(t) \rightarrow 0$; leaving the forced response only.

This case of τ +ve is said to be stable

⇒ When τ is -ve natural response grows beyond the bound as $t \rightarrow \infty \Rightarrow x_n(t) \rightarrow \infty$

The forced response become negligible compare to natural response & thus at this case of natural response is said to be unstable.

I order circuits of OP AMPS and dependent source may be unstable

Response of 1 order ckt to a Non-constant source.

General form of 1 order circuits diff eqⁿ for RL or RC circuit is

$$\frac{dx(t)}{dt} + ax(t) = y(t) \quad \text{--- (1)}$$

↳ Input / Forcing system

When $y(t)$ is a const only when we have constant current or constant voltage source & $a = \frac{1}{\tau}$ is the reciprocal of time const

Now consider eqⁿ (1) as given by $\frac{d}{dt}x(t) + ax(t) = y(t)$ can be written as derivative of product of two terms.

$$\frac{d}{dt}(xe^{at}) = \frac{d}{dt}x e^{at} + ax e^{at} \quad \text{--- (A)}$$

$$\frac{d}{dt}x e^{at} + ax e^{at} = \left[\frac{dx}{dt} + ax \right] e^{at} \quad \text{--- (2)}$$

Sub (1) & (2)

$$\frac{d}{dt}(xe^{at}) = ye^{at} \quad [\text{by (1)}]$$

$$\frac{d}{dt}(xe^{at}) = ye^{at} \quad [\text{By (A)}]$$

$$\int \frac{d}{dt}(xe^{at}) = \int ye^{at}$$

$$xe^{at} = \int ye^{at} dt + K$$

$$x = \underbrace{e^{-at} \int ye^{at} dt}_{x_f} + \underbrace{e^{-at} K}_{x_N}$$

If $y(t) = M$ (const)

$$x(t) = \underbrace{\frac{M}{a}}_{x_f} + \underbrace{Ke^{-at}}_{x_N}$$

Now consider the case when $y(t)$ or the forcing function is not a const.

If $y(t) = e^{bt}$

$$x_f = e^{-at} \int e^{bt} \cdot e^{at} dt$$

$$x_f = \frac{e^{bt}}{a+b} ; a+b \neq 0$$

$$\Rightarrow x_f = K$$

$$y(t) = M$$

$$x_f = e^{bt}$$

$$= e^{-at}$$

$$= Me^{-at}$$

As it can be seen from the rule of integration, we get $x(t)$

$$x_f(t) =$$

Now then in eqⁿ (1)

Summary

forced response

$$x(t) =$$

$$\frac{y(t)}{a}$$

$$\textcircled{1} y(t) = M$$

$$\textcircled{2} y(t) = e^{bt}$$

$$\textcircled{3} y(t) = M \sin \omega t$$

$$\textcircled{4} y(t) = M \sin \omega t$$

$$\Rightarrow x_f = K e^{bt} \quad x_N = K e^{-t/\tau}; a = \frac{1}{\tau}$$

$$y(t) = M \sin(\omega t + \theta)$$

$$x_f = e^{-at} \int y(t) e^{at} dt$$

$$= e^{-at} \int M \sin(\omega t + \theta) e^{at} dt$$

$$= M e^{-at} \int \sin(\omega t + \theta) e^{at} dt \rightarrow \text{At this Integrating}$$

As it can be seen the forced response now is to be solved by using I.L.A.T.E rule of integration for product of two functions so it is hence conv. to get solⁿ of x_f as

$$x_f(t) = A \sin \omega t + B \cos \omega t \quad \text{where } A \text{ \& } B \text{ are const.}$$

Now these const can be found out by substituting value of $x_f(t)$

in eqn ①

Summary

forced f^n depends upon value of response applied

$$x(t) = x_N(t) + x_f(t)$$

$$\frac{y(t)}{x(t)}$$

$$\textcircled{1} y(t) = M$$

$$\textcircled{2} y(t) = M e^{bt}$$

$$\textcircled{3} y(t) = M e^{-bt}$$

$$\textcircled{4} y(t) = M \sin(\omega t + \theta)$$

$$\frac{x_f(t)}{x(t)}$$

$$x_f = N$$

$$x_f = N e^{bt}$$

$$x_f = N e^{-bt}$$

$$x_f = A \sin \omega t + B \cos \omega t$$

$$y(t) = e^{-bt}$$

$$x_f = \frac{e^{-bt}}{a-b}$$

They

This form of response is indeterminate for this case we consider force response as ~~if~~

$$x_f = t e^{-bt}$$

This is the condition when natural response already has a term of same power or same form as the forcing function

Time constant

$$i(t) = \frac{V_0}{R} + \frac{L}{R} \left[\frac{di}{dt} \right] e^{-t/\tau} \quad \left\{ \tau = L/R \right\}$$

$$\Rightarrow i(t) = I_0 + [i(0) - I_0] e^{-t/\tau}$$

$$\text{if } i(0) = 0$$

$$i(t) = \frac{V_0}{R} [1 - e^{-t/\tau}]$$

$$\text{at } t = \tau$$

$$i(\tau) = \frac{V_0}{R} [1 - 1/e]$$

$$= \frac{V_0}{R} 0.63$$

$$\frac{i(\tau)}{V_0/R} = 0.63$$

$$V_0/R$$

$$\frac{i(\tau)}{I_0} = 0.63$$

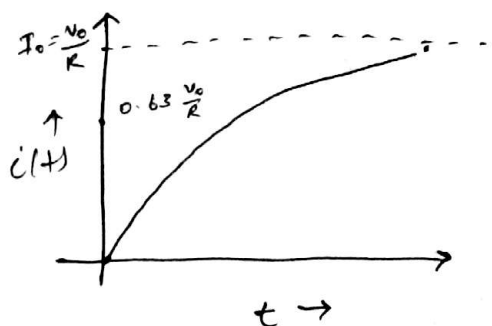
$$I_0$$

↓

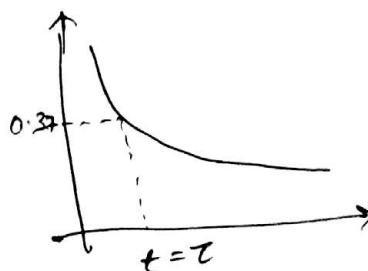
Initial value of current or steady state value

En this τ is defined as time required to reach 63% of its final value of current. This is the condition when IC is given

when IC = 0

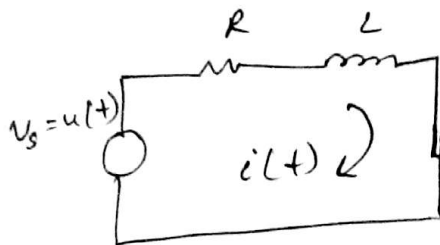


when IC = finite



Find time constant for RL circuit

Step Response of RL circuit



$$-v_s + Ri(t) + L \frac{di}{dt} = 0$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{u(t)}{L}$$

Comparing

$$\frac{dx(t)}{dt} + ax(t) = y(t)$$

$$x_f = e^{-at} \int y(t) e^{at} dt$$

$$= \frac{e^{-at}}{L} \int e^{at}$$

$$[\because u(t) = 1]$$

$$= \frac{e^{-at}}{L} \cdot \frac{e^{at}}{a} = \frac{e^{-at}}{L} \cdot \frac{e^{at}}{a}$$

$$= \frac{e^{-at}}{L}$$

$$i_f = x_f = \frac{1}{L} \quad [\because \frac{1}{a} = \tau]$$

$$K e^{-t/\tau} + \frac{1}{L} = i(t)$$

Assuming $i(0) = 0$

$$K + \frac{1}{L} = 0$$

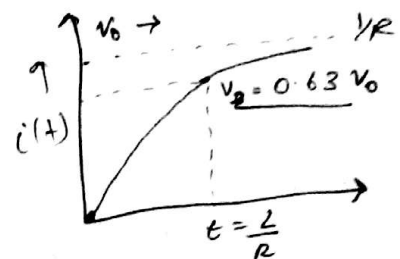
$$K = -\frac{1}{L}$$

$$i(t) = -\frac{1}{L} e^{-t/\tau} + \frac{1}{L}$$

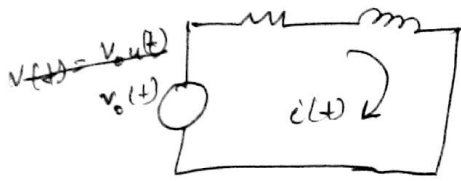
$$= \frac{1}{L} [1 - e^{-t/\tau}]$$

$$= \frac{L}{RL} [1 - e^{-tR/L}]$$

$$i(t) = \frac{1}{R} [1 - e^{-tR/L}]$$



Pulse Response of 1st order



$$v(t) = v_0 u(t) - v_0 u(t-t_0)$$

$$i(0) = 0$$

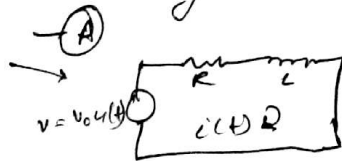
Using super position

→ calculating $i(t)$ for $v_0 u(t)$

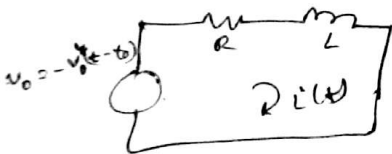
→ then " " " $v_0 u(t-t_0)$

→ Adding them up

$$i(t) = \frac{v_0}{R} [1 - e^{-t/\tau}]$$



for



$$i(t) = \frac{v_0}{R} [i(0) - \frac{v_0}{R}] e^{-t/\tau}$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{v_s}{L}$$

$$= \frac{v_0 u(t-t_0)}{R}$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{v_0 u(t-t_0)}{L}$$

$$i_f = e^{-at} \int \frac{v_0}{L}$$

$$i(t) = -\frac{v_0 u(t-t_0)}{R} [i(0) - \frac{v_0 u(t-t_0)}{R}] e^{-(t-t_0)/\tau}$$

$$= \frac{v_0 u(t-t_0)}{R} [1 - e^{-(t-t_0)/\tau}]$$

$$= \frac{v_s}{R} [1 - e^{-(t-t_0)/\tau}] \quad \text{--- (B)}$$

lamp response = (A) + (B)

$$i(t) = \frac{V_0}{R} [1 - e^{-t/\tau}] - \frac{V_0}{R} [1 - e^{-(t-t_0)/\tau}]$$

$$i(t) = \frac{V_0}{R} e^{-t/\tau} [e^{t_0/\tau} - 1] \rightarrow \text{pulse excitation}$$

at $t = t_0$

$$i(t) = \frac{V_0}{R} [1 - e^{-t_0/\tau}]$$

Waveform for Fig (1) is obtained by (A) & (B)

$$i_1(t) = \frac{V_0}{R} [1 - e^{-t/\tau}] \rightarrow (A)$$

$$i_2(t) = -\frac{V_0}{R} [1 - e^{-(t-t_0)/\tau}]$$

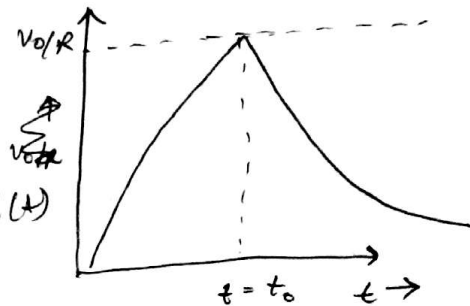
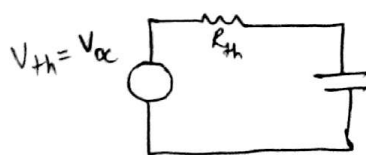
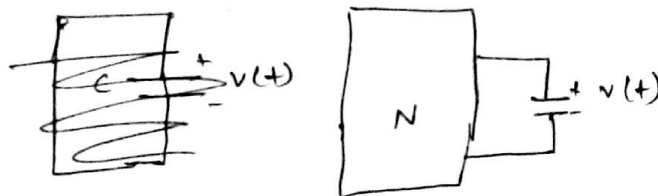


Fig (1)

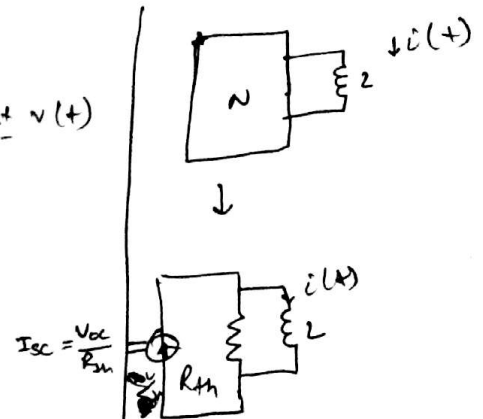
Important Points

* 1



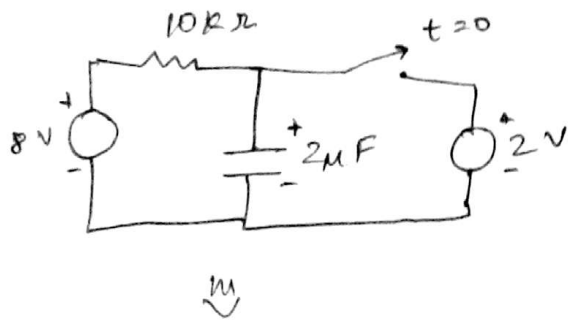
$$v(t) = V_{oc} + [v(0) - V_{oc}] e^{-t/\tau}$$

$$\tau = R_{th} C$$

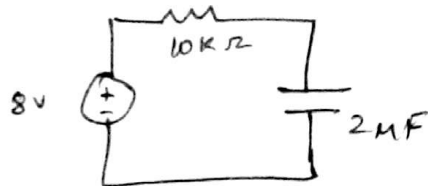


$$i(t) = I_{sc} + (i(0) - I_{sc}) e^{-t/\tau}$$

Find capacitor voltage after the switch opens in circuit as shown what is the value of cap. voltage after some time after the switch opens



as the switch is opening at $t=0$ $\therefore$ it is assumed it is in steady state $\therefore V_0 = 2V$



Do again with Force & natural

$$v(t) = V_0 + [V_0 - V_0] e^{-t/\tau}$$

$$= 8 + [2 - 8] e^{-t/\tau}$$

$$v(t) = 8 - 6e^{-t/\tau}$$

$$v(t) = 8 - 6e^{-t/\tau}$$

$$= 8 - 6e^{-t/\tau}$$

$$= 8 - 6e^{-t/\tau}$$

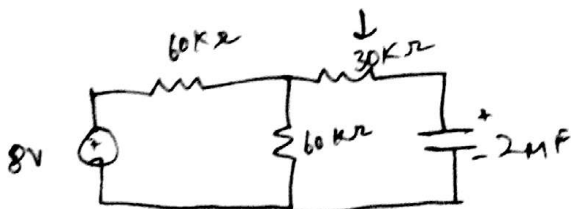
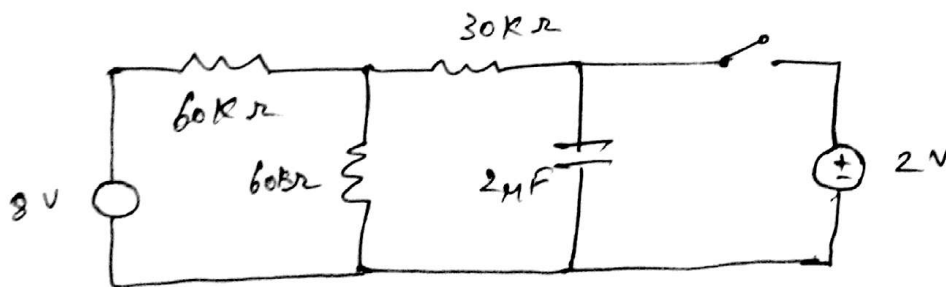
$$= 8 - 6e^{-t/\tau}$$

$$= 8 - 6e^{-t/20 \times 10^{-3}}$$

$$\tau = RC = 20 \text{ ms}$$

The circuit in figure is at steady state before switch opens find the current at $t > 0$

$$V_0 = 2V$$

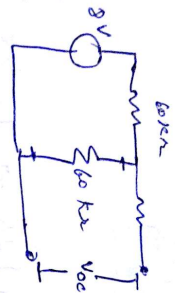


4V
8V
the

$R_{th} = 60\text{K}\Omega$ [$2 \times 60\text{K}\Omega$ in // & one $30\text{K}\Omega$ in series]

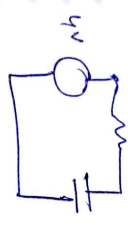
$V_{oc} =$ Voltage across $60\text{K}\Omega$ branch

The node of $30\text{K}\Omega$ is gone as no current

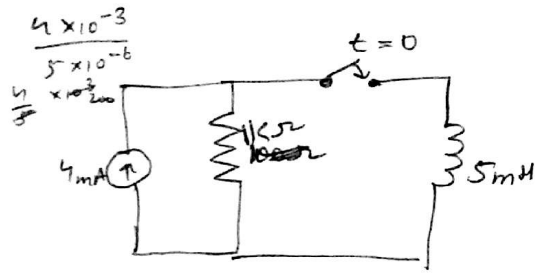


$$V_{oc} = \frac{60}{60+60} \times 8$$

$$V_{oc} = 4\text{V}$$



Q. Find the inductor current after the switch closes in the circuit shown below
How long will it take for L to reach 20 mA value



$$\tau = \frac{L}{R} = \frac{5 \times 10^{-3}}{10^3}$$

$$\tau = 5 \text{ ms}$$

$$V = IR$$

$$= 4 \times 10^{-3} \times 10^3$$

$$V = 4 \text{ V}$$

$$i_s = \frac{V_0}{R} [1 - e^{-t/\tau}]$$

$$i = \frac{4}{1000}$$

$$i = 4 \text{ mA}$$

$$i(t) = I_{sc} + [I(0) - I_{sc}] e^{-t/\tau}$$

$$= 4 + [0 - 4] e^{-t/5} \text{ mA}$$

$$i(t) = 4 - 4e^{-t/5}$$

~~20 ms~~

OR

$$i_N = K e^{-t/\tau}$$

$i_f = N$ [As source is constant forced response will be constant]

$$i(t) = K e^{-t/\tau} + N$$

$$i_f = e^{-at} \int y(t) e^{at} dt$$

$$= e^{-at} \int \left[\frac{R}{L} I_{sc} \right] e^{at} dt$$

$$= 4e^{-at} \int a e^{at} dt$$

$$= 4e^{-at} \cdot \frac{1}{a} e^{at}$$

$$i_f = 4$$

$$\Rightarrow i(t) = K e^{-t/\tau} + 4$$

$$\text{Now } i(0) = 0$$

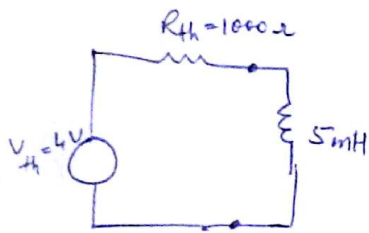
$$i(0) = K + 4$$

$$K = -4$$

$$i(t) = 4 - 4e^{-t/\tau}$$

or

By charging in Thevenin Circuit



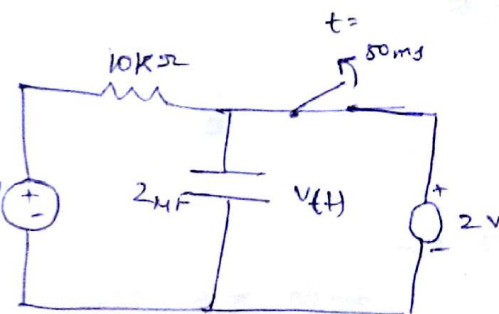
$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = \frac{V_0}{L}$$

$$i(t) = \frac{V_0}{R} + \left[i(0) - \frac{V_0}{R} \right] e^{-t/\tau}$$

$$= \frac{4}{1000} + \left[0 - \frac{4}{1000} \right] e^{-t/15}$$

$$i(t) = 4 - 4 e^{-t/15} \text{ mA}$$

Q Find capacitor voltage after the switch opens in circuit as shown below what is the value of Capacitor Voltage 50ms after it opens.



↓ After opening at $t = 0.05s$



$$\tau = RC = 10 \times 10^3 \times 2 \times 10^{-6}$$

$$= 2 \times 10^{-2}$$

$$\approx 0.02 = 20 \text{ ms}$$

$$V(t) = V_0 + [V(0.05) - V_0] e^{-(t-50)/20}$$

$$= V_0 + [V(0.05) - V_0] e^{-}$$

$$= 6 + [V(0.05) - 6] e^{-}$$

$$V(0.05) = 2 \text{ V (As max it can charge to 2V)}$$

$$V(t) = V_0 + [V(0.05) - V_0] e^{-t/\tau}$$

$$= 8 - 6 e^{-\frac{(t-50)}{20}} \text{ V}$$

OR

$$V_N = K e^{-(t-t_0)/\tau}$$

$$V_f = N$$

$$V_f = e^{-at} \int y(t) e^{at} dt$$
$$= e^{-at} \int \frac{8}{20 \times 10^{-3}} e^{at} dt$$

$$\boxed{\frac{V_f}{e^{-at}} = 8}$$

$$V_N = K e^{-t/\tau}$$

$$V(t) = K e^{-t/\tau} + 8$$

$$V(0) = K e^{0/\tau} + 8$$

$$2 = K + 8$$

$$\boxed{K = -6}$$

$$V(t) = 8 - 6 e^{-t/\tau}$$

$$V(t) = K e^{-(t-t_0)/\tau} + 8$$

$$V(t) = K e^{-(t-50)/\tau} + 8$$

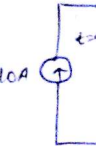
at 50ms

$$V(50) = K e^{-(50-50)/\tau} + 8$$

$$2 = K + 8$$

$$\boxed{K = -6}$$

Q2 I_0

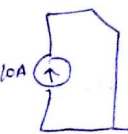


at t



i(t)

at t = 0



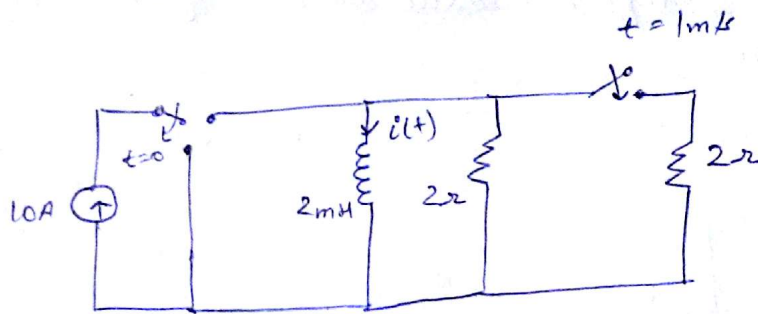
at

i(t)

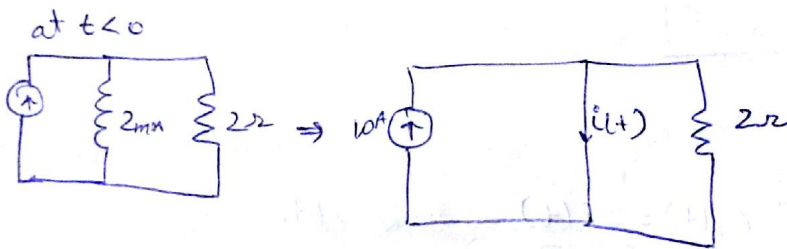
i(1) =

at t

Q3 ~~12~~

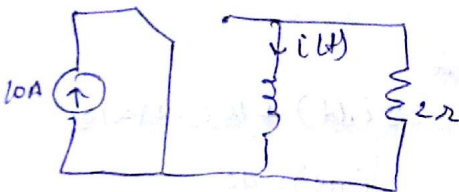


Suppose the ckt. reaches steady state at time $t=0$ so that current across inductor at $t=0$ will become IC current.



$i(0^-) = i(0^+) = 10A$ [Current across inductor does not change instantaneously]

at $t=0^+$



$$\tau = \frac{2mH}{2\Omega} = 1ms$$

$$i(t) = 10e^{-t}$$

$$i(t) = I_{sc} + [I(0) - I_{sc}]e^{-t/\tau}$$

$$= 0 + [10 - 0]e^{-t/\tau}$$

$$= 10e^{-t/\tau}$$

Current from current source = 0

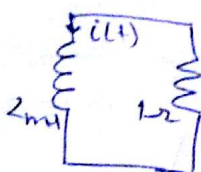
at $1ms$

$$i(1) = 10e^{-1/1}$$

as $\tau=1$

$$i(1) = 3.68A$$

at $t > 1ms$



$$i(1^-) = i(1^+) = 3.68$$

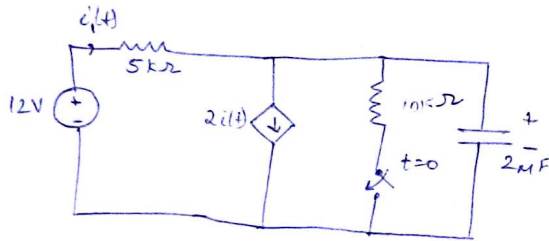
$$i(t) = I_{sc} + [i(1) - I_{sc}]e^{-(t-1)/\tau}$$

current from current source = 0

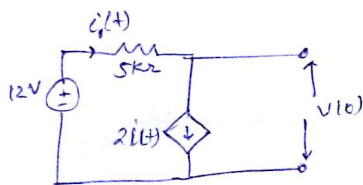
$$i(t) = 0 + 3.68e^{-(t-1)/\tau}$$

$$i(t) = 3.68e^{-(t-1)/2}$$

Q The first order ckt shown below is at steady state before the switch closes at $t=0$ the ckt contain a dependent source and so it may be unstable find the cap. voltage V_c for $t > 0$

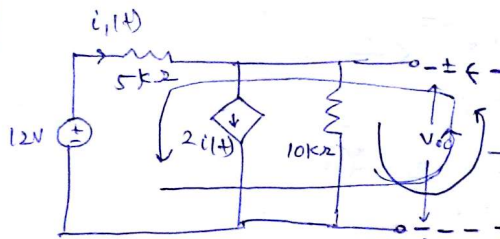


$t < 0$



$i_1(t) = 2i(t)$ as series ckt.
 $V(0) = 12V$ is IC

$\frac{12}{5}$



$i_1(t) = 5 \times \frac{1}{10} + 2$
 $V = IR$
 $= \frac{12}{5} \times \frac{10}{10+5}$
 $V = \frac{24}{5}$

$5i_1(t) + 10i_1(t) = 12$

$i_1(t) = \frac{12}{5}$

$i_1(t) = 2.4A$

applied 1V
 $V = \frac{I \cdot 10}{I + 10k}$

$-1 + 5(-i(t)) = 0$

$R_{th} = \frac{1}{I}$

$I_1 = \frac{1}{10k} = 10 \mu A$

$i(t) = -\frac{1}{5} mA$

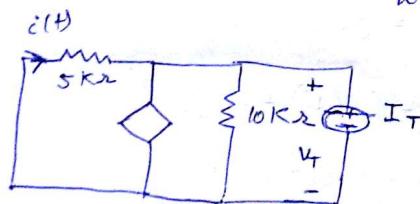
$[I - i(t)] 10 - 1 = 0$

$I = \frac{1}{10} + i(t)$
 $I = -\frac{1}{10} mA$

$-i(t) + I = \frac{1}{10}$

$0 = (5 \times 10^3) i(t) + (10 \times 10^3) \times (I + \frac{1}{10} - 2i(t))$

$i_1 = 2i(t)$



$V_{oc} = 24V$

$V(t) = 24 - 12e^{-t/20} V$

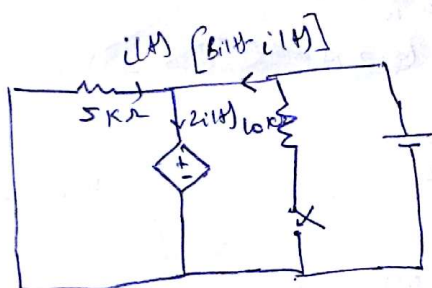
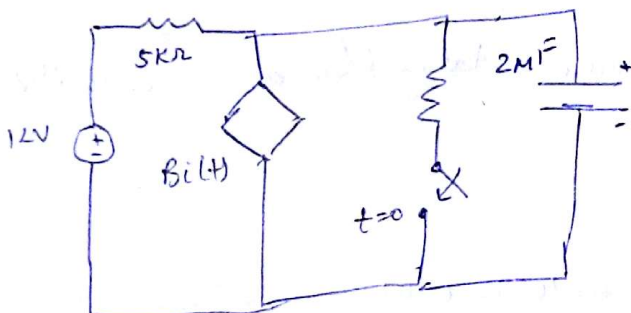
$R_{th} = \frac{-10 \times 10^3 \times I_T}{I_T}$
 $R_{th} = \frac{-1}{I_T}$
 $= -10k \Omega$
 $V(t) = V_{oc} + (V(0) - V_{oc})e^{-t/\tau}$

stable

Q The ckt considered in prev example has been redrawn as shown below with a gain of dependent source rep. by variable B . what restriction must be placed on the gain of dependent source to ensure ckt is stable.

given $\tau = 20 \text{ ms}$

$\tau = R_{th} C$



for big loop
 $i(t) = 1/5 \text{ mA}$

for smaller

$$-1 \left[I + (B i(t) - i(t)) \right] R = 0$$

$$-1 \left[I + \frac{1}{5} - \frac{B}{5} \right] R = 0$$

$$I = \frac{3-2B}{10} \text{ mA}$$

$$I = \frac{B i(t) - 1}{5}$$

$$I = 1 - \frac{B}{5}$$

$$R_{th} = \frac{10 \times 10^3}{3-2B}$$

for $\tau > +ve$
 $R_{th} = +ve$ as C is already +ve

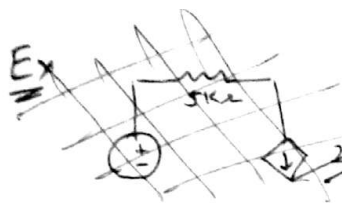
only possible $3-2B > 0$
 $B < 1.5$

Now $\tau = R_{th} C = 20 \times 10^{-3}$
 ~~$\rightarrow 20 \times 10^{-3}$~~

$$\frac{10 \times 10^3}{3-2B} \times 2 \times 10^{-4} = 20 \times 10^{-3}$$

$$3-2B = 1$$

$$\boxed{B=1}$$



$$v(t) = 4 - 8u(t)$$

$$i_0(t) = ?$$

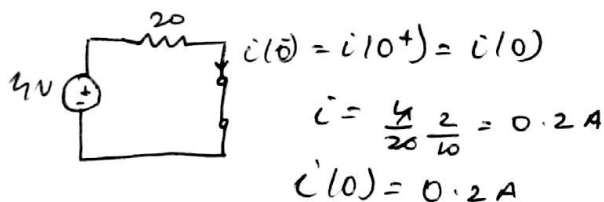
$$I(t) = \frac{V_0}{R} + \left[i(0) - \frac{V_0}{R} \right] e^{-t/\tau}$$

$$v_s = 4 - 8u(t)$$

$$\begin{array}{l|l} t < 0 & t > 0 \\ v_s = 4V & v_s = -4V \end{array}$$

at $t=0$ disturbance is taking place as same as switch

at $t=0^-$

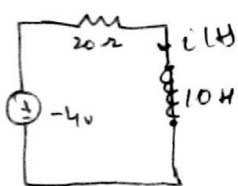


$$i(0^-) = i(0^+) = i(0)$$

$$i = \frac{4}{20} = 0.2A$$

$$i(0) = 0.2A$$

at $t=0^+$



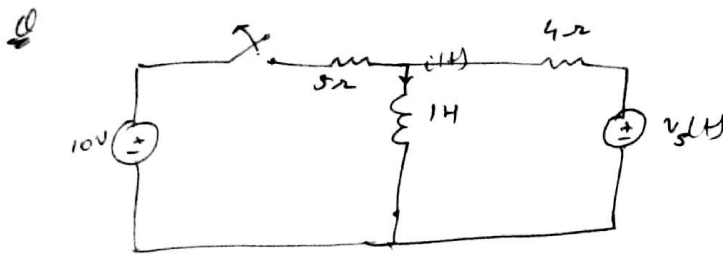
$$I(t) = \frac{V_0}{R} + \left[i(0) - \frac{V_0}{R} \right] e^{-t/\tau}$$

$$= \frac{-4}{20} + \left[0.2 + \frac{4}{20} \right] e^{-t/0.5}$$

$$= -0.2 + 0.4e^{-2t}$$

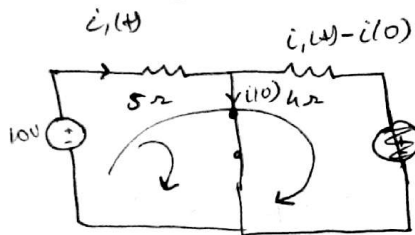
$$i(t) = \begin{cases} 0.2, & t < 0 \\ -0.2 + 0.4e^{-2t}, & t > 0 \end{cases}$$

Find $i(t)$ if $v_s = 10e^{-2t}u(t)$
 assume the ckt is in steady
 state condition at $t = 0^-$



at $t \leq 0$ $v_s = 10e^{+2t}$ | at $t > 0$ $v_s = 10e^{-2t}$

at $t = 0^-$



$$i(0^-) = i(0^+) = i(0) = \frac{10}{5 + 4} = \frac{10}{9} = 1.11 \text{ A}$$

$$= \frac{10}{9}$$

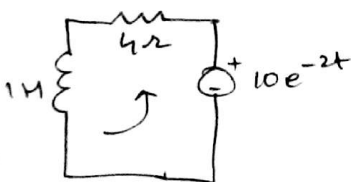
$$-10 + 5i_1(t) = 0$$

$$i_1(t) = 2 \text{ A}$$

$$-10 + 5i_1(t) + [i_1(t) - i(0)]4 = 0$$

$$i(0) = 2 \text{ A}$$

at $t = 0^+$



$$\text{let } i_f = Be^{-2t}$$

$$-10e^{-2t} + 4i(t) + L \frac{di(t)}{dt} = 0$$

$$4i(t) + \frac{di(t)}{dt} = 10e^{-2t}$$

$$\tau = \frac{L}{R} = 0.25 \text{ s}$$

$$4Be^{-2t} + B(-2)e^{-2t} = 10e^{-2t}$$

$$\begin{aligned} 5B &= 10 \\ B &= 2 \end{aligned} \quad \begin{aligned} 2B &= 10 \\ B &= 5 \end{aligned}$$

$$\therefore i_f = 5e^{-2t}$$

$$\begin{aligned} i_N &= Ke^{-t/\tau} \\ &= Ke^{-t/0.25} \end{aligned}$$

$$i_N = Ke^{-4t}$$

$$\begin{aligned} i(t) &= i_N + i_f \\ i(t) &= Ke^{-4t} + 5e^{-2t} \end{aligned}$$

$$i(0) = 2A$$

$$2 = K + 3$$

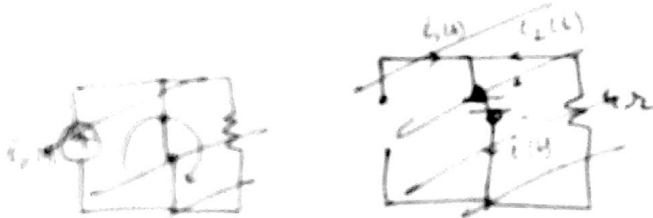
$$K = -3$$

$$i(t) = 5e^{-2t} - 3e^{-4t}$$

Q Find $v(t)$ for $t > 0$ if $i_s = 10 \sin 2t u(t)$ A and $v(0) = 0$ for det

$$i_f = A \sin 2t + B \cos 2t$$

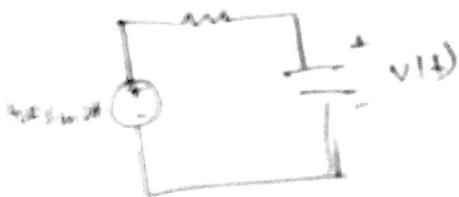
$$v(t) = i_s(t) + \frac{1}{C} \int i dt$$



$$V(t) = V_{oc} + (V(0) - V_{oc}) e^{-t/\tau}$$

$$V(t) = V_{oc} + (V(0) - V_{oc}) e^{-t/\tau}$$

$$\tau = RC = 4 \times \frac{1}{2} = 2s$$



$$v(t) =$$

$$-40 \sin 2t + 4 i_c(t) + v(t) = 0$$

$$-40 \sin 2t + 4 \left(\frac{dv(t)}{dt} \right) + v(t) = 0$$

$$-40 \sin 2t + 4 \left(\frac{dv(t)}{dt} \right) + v(t) = 0$$

$$2 \frac{dv(t)}{dt} + v(t) = 40 \sin 2t$$

$$v(t) = V_c + V_{oc}$$

$$v_f = A \sin 2t + B \cos 2t$$

$$A \sin 2t + B \cos 2t \left[A \cos 2t - 2B \sin 2t \right] + A \sin 2t + B \cos 2t = 40 \sin 2t$$

$$4A \cos 2t - 4B \sin 2t + A \sin 2t + B \cos 2t = 40 \sin 2t$$

$$6 \cos 2t (4A + B) + 5 \sin 2t (-4B + A) = 40 \sin 2t$$

$$A - 4B = 40 \times 4$$

$$4A + B = 0$$

$$4A - 16B = 160$$

$$4A + B = 0$$

$$\begin{array}{r} - \\ - \\ \hline -17B = 160 \end{array}$$

$$B = -\frac{160}{17}$$

$$A = \frac{160 + 16B}{4}$$

$$A = \frac{40}{17}$$

$$V_f = \frac{40}{17} \sin 2t - \frac{160}{17} \cos 2t$$

$$V_n = K e^{-t/\tau}$$

$$V(t) = K e^{-t/\tau} + \frac{40}{17} \sin 2t - \frac{160}{17} \cos 2t$$

$$\text{at } t=0; V(0)=0$$

$$0 = K - \frac{160}{17}$$

$$K = \frac{160}{17}$$

$$V(t) = \frac{160}{17} e^{-t/\tau} + \frac{40}{17} \sin 2t - \frac{160}{17} \cos 2t$$